INCH-POUND

MIL-DTL-83739/1B
15 November 1999
SUPERSEDING
MIL-S-83739/1A
3 November 1975

DETAIL SPECIFICATION SHEET

# SWITCH, ANTENNA, RADIO FREQUENCY, SOLID-STATE, SA-1474A/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switch described herein shall consist of this document and the latest issue of MIL-DTL-83739.

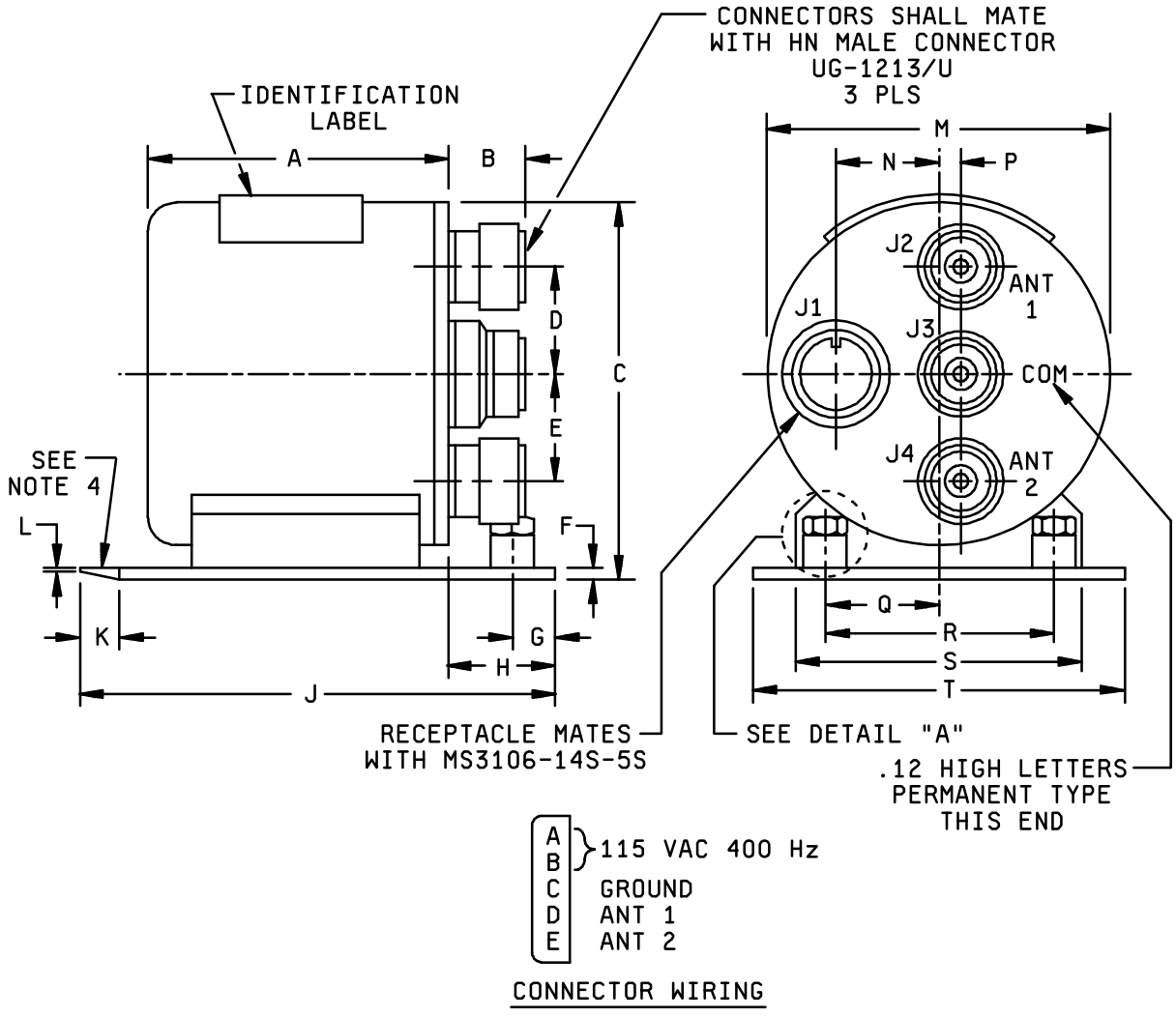


FIGURE 1. Dimensions and configuration.


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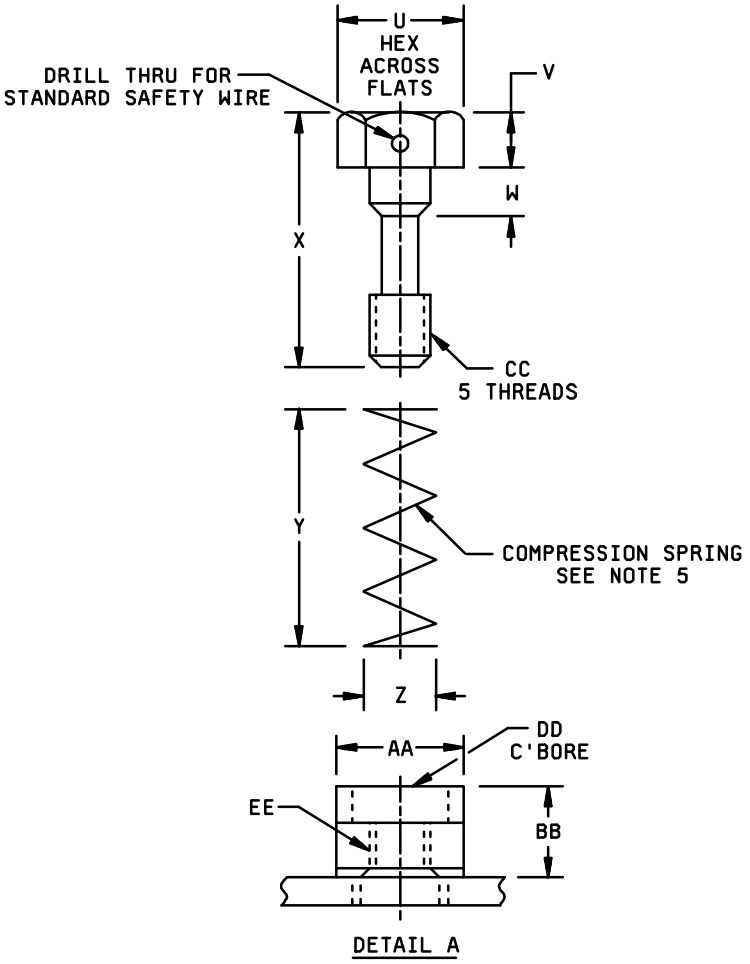


| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | 2.57 | 2.69 | 65.3 | 68.3 |
| B | .64 | .70 | 16.3 | 17.8 |
| C | --- | 3.44 | --- | 87.4 |
| D | .927 | .947 | 23.55 | 24.05 |
| E | .927 | .947 | 23.55 | 24.05 |
| F | .090 | .110 | 2.29 | 2.79 |
| G | .365 | .385 | 9.27 | 9.78 |
| H | .925 | .945 | 23.50 | 24.00 |
| J | 4.13 | 4.19 | 104.9 | 106.4 |
| K | .334 | .354 | 8.48 | 8.99 |
| L | .006 | .026 | 0.15 | 0.66 |
| M | 2.97 | 3.03 | 75.4 | 77.0 |
| N | .87 | .93 | 22.1 | 23.6 |
| P | .16 | .22 | 4.1 | 5.6 |
| Q | .97 | 1.03 | 24.6 | 26.2 |
| R | 1.97 | 2.03 | 50.0 | 51.6 |
| S | 2.47 | 2.53 | 62.7 | 64.3 |
| T | 3.22 | 3.28 | 81.8 | 83.3 |
| U | .425 | .445 | 10.80 | 11.30 |
| V | .178 | .198 | 4.52 | 5.03 |
| W | .115 | .135 | 2.92 | 3.43 |
| X | .865 | .885 | 21.97 | 22.48 |
| Y | .615 | .635 | 15.62 | 16.13 |
| Z | .240 | .260 | 6.10 | 6.60 |
| AA | .428 | .448 | 10.87 | 11.38 |
| BB | .302 | .322 | 7.67 | 8.18 |
| CC | .190-24 UNC 2A | | 4.83-24 UNC-2A | |
| DD | .312 DIA x .125 DEEP | | 7.92 DIA x 3.18 DEEP | |
| EE | .190-24 UNC 2B | | 4.83-24 UNC-2B | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.03 inch (0.76 mm) for two place decimals and ± .010 inch (0.25 mm) for three place decimals.
4. The upper and lower surface of this area is not to be painted.
5. The spring shall be strong enough to hold the bolt free when disengaged from mounting plate.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 2 pounds maximum.

Finish: Black paint.

Mounting: Mounting provisions and mounting holes shall be provided that will interface with mounting plate MT-1948/U covered by AF Drawing 59C12539, NSN 5985-00-788-6960).

Configuration: Single pole double throw (1P2T).

Sequence: Make-before-break.

Remote control: Provision shall be provided to remotely select either RF output on a full time basis by grounding pins on the power connector. Pin D shall be grounded for remote selection of Ant 1. Pin E shall be grounded for remote selection of Ant 2. No damage shall be caused if pins D and E are grounded simultaneously.

Nominal impedance: 50 ohms.

Frequency range: 1.02 to 1.1 GHz.

Housing: Type E.

Switching rate: 38 Hz ± 5 percent.

Primary input power: 10 watts max, with the input voltage and frequency at 115 ± 7 volts and 400 ± 20 Hz, respectively.

Transition time: Less than 25 microseconds.

Voltage standing wave ratio (VSWR):

Either output to common terminal: 1.2:1.

Transition time: 2.5:1.

Isolation (crosstalk): 30 dB minimum between output.

Insertion loss: 0.5 dB maximum.

RF power handling capability: 1 kilowatt peak pulse power with 1 percent duty cycle into a mismatched load with a VSWR rating of 2:1.

MTBF: Design for 10,000 hours, with a guaranteed minimum of 5,000 hours.

Temperatures:

Operating: -55°C to +121°C.

Shock: -40°C to +121°C.

Part or Identifying Number (PIN): M83739/1-01.

NOTE: The switch covered by this specification sheet is substitutable for the following commercial types:

| Manufacturer's code | Commercial type | FSN 5985- |
|---|---|---|
| 82152 | 70D10100 | 932-4257 |
| 74868 | 351-12010 | |
| 96412 | DODA1MS66-358 | 932-4257 |
| 80058 | SA1474/A | 932-4257 |
| 22887 | 001-1012858 | 932-4257 |



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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-1205-01)



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